



Lab Report

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Team: E1

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Goal

1. Connect the wires between Arduino, breadboard, motor controller, and battery, so the vehicle can be able to follow the lane.

Discussion

1. We uploaded the code that worked fine in Tinkercad to the Arduino, wired everything up, connected the battery, and flipped the switch, but the vehicle didn't respond at all.

After repeated checks, we found:

1.1 The breadboard was wired incorrectly, and some parts weren't receiving power.

1.2 The positive and negative terminals of the motor were connected incorrectly.

2. After multiple adjustments, the motor finally spun normally, but we couldn't control its speed. So, we checked again and found:

2.1 We realized that not all digital pins can output PWM. So we looked for the tilde symbol (~) next to the pin number on Arduino (e.g., pins 3, 5, 6, 9, 10, and 11), and adjusted the wires.

Finally, the vehicle was able to move slowly. For details, please refer to the video and codes in the folder.

3. Currently, our vehicle cannot detect black lines or obstacles. We believe this may be because the ultrasonic sensors and IR sensors are interfering with each other.

We will adjust the code to use the IR sensors to detect the lane first, and then use the ultrasonic sensors to detect obstacles.

